

Loan Default Analysis Project Documentation

This document contains the complete Exploratory Data Analysis (EDA), statistical concepts, probability analysis, visualizations, interpretations, and final conclusion for the Loan Default Analysis Project.

Step 1 — Import Libraries

Import pandas, numpy, matplotlib, seaborn, scipy.

Interpretation: Used for data analysis, visualization, and statistics.

Step 2 — Upload Dataset

Upload dataset into Google Colab.

Interpretation: Allows the dataset to be accessed for analysis.

Step 3 — Load Dataset

Load CSV using pandas.

Interpretation: Converts data into DataFrame format.

Step 4 — Display Dataset

Use head() and tail().

Interpretation: Displays first and last records.

Step 5 — Dataset Shape

Use df.shape.

Interpretation: Shows rows and columns.

Step 6 — Dataset Information

Use df.info().

Interpretation: Shows datatype and null values.

Step 7 — Missing Values

Use df.isnull().sum().

Interpretation: Detects missing data.

Step 8 — Statistical Summary

Use df.describe().

Interpretation: Shows mean, std, min, max, quartiles.

Step 9 — Mean, Median, Mode

Calculate central tendency.

Interpretation: Helps analyze average and typical values.

Step 10 — Range, Variance, Standard Deviation

Calculate data spread.

Interpretation: Measures variability and financial risk.

Step 11 — Correlation Matrix

Use `corr()`.

Interpretation: Identifies variable relationships.

Step 12 — Heatmap

Visualize correlations.

Interpretation: Detects positive and negative relationships.

Step 13 — Histogram

Plot distributions.

Interpretation: Shows income and loan patterns.

Step 14 — Boxplot

Detect outliers.

Interpretation: Identifies abnormal financial values.

Step 15 — Scatter Plot

Income vs Loan Amount.

Interpretation: Analyzes customer borrowing behavior.

Step 16 — Countplot and Pie Chart

Visualize defaults.

Interpretation: Shows default distribution.

Step 17 — Pairplot

Visualize all numerical relations.

Interpretation: Detects trends and patterns.

Step 18 — Skewness and Kurtosis

Measure distribution shape.

Interpretation: Analyzes asymmetry and outliers.

Step 19 — Gaussian Curve

Check normal distribution.

Interpretation: Compares actual and normal distribution.

Step 20 — Default Probability

Calculate loan default probability.

Interpretation: Measures customer risk.

Step 21 — Conditional Probability

Probability for low credit customers.

Interpretation: Identifies risky customer groups.

Step 22 — Contingency Table

Use `crosstab()`.

Interpretation: Analyzes categorical relationships.

Step 23 — Chi-Square Test

Statistical relationship testing.

Interpretation: Checks variable dependency.

Step 24 — Dot Product

Apply vector multiplication.

Interpretation: Measures vector similarity.

Step 25 — Vector Norm

Calculate vector magnitude.

Interpretation: Measures financial vector strength.

Step 26 — Angle Between Vectors

Calculate cosine angle.

Interpretation: Measures similarity direction.

Step 27 — Final Conclusion

Summarize project findings.

Interpretation: Supports banking risk management.

Important Formulas

- Mean = $\Sigma x / n$
- Range = Maximum – Minimum
- Variance = $\Sigma(x-\mu)^2 / N$
- Standard Deviation = $\sqrt{[\Sigma(x-\mu)^2 / N]}$
- Conditional Probability = $P(A|B) = P(A \cap B) / P(B)$
- Bayes Theorem = $P(A|B) = P(B|A)P(A) / P(B)$
- Default Probability = Default Customers / Total Customers
- $\cos\theta = (A \cdot B) / (||A|| ||B||)$

Final Conclusion

The Loan Default Analysis Project successfully analyzed customer financial behavior using statistics, probability, visualization, and linear algebra concepts. The project identified important factors affecting loan default such as credit score, income, and loan amount. The analysis helps banks identify risky customers, improve loan approval decisions, reduce financial loss, and build future machine learning models.